Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 105610
Subject: Soft Computing
Time: 03 Hours
Full Marks: 70

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) Soft computing is primarily concerned with:
- (i) Deterministic algorithms
- (ii) Exact mathematical models
- [✓ (iii) Approximate and flexible reasoning]
- (iv) Binary logic

(b) Which of the following is *not* a constituent of soft computing?
- (i) Fuzzy Logic
- (ii) Neural Networks
- (iii) Genetic Algorithms
- [✓ (iv) Compiler Design]

(c) Soft computing is inspired by:
- (i) Boolean logic
- [✓ (ii) Human reasoning and biological systems]
- (iii) Classical mathematics
- (iv) Symbolic logic

(d) Computational Intelligence (CI) includes:
- [✓ (i) Neural networks, fuzzy systems, and evolutionary algorithms]
- (ii) Operating systems and compilers
- (iii) Data structures and algorithms
- (iv) Object-oriented programming

(e) Reinforcement learning is based on:
- (i) Error backpropagation
- [✓ (ii) Reward and punishment signals]
- (iii) Random updates
- (iv) Gradient descent

(f) Fuzzy logic is based on:
- (i) Binary truth values
- [✓ (ii) Multi-valued or approximate reasoning]
- (iii) Boolean algebra
- (iv) Symbolic rules

(g) Fuzzy-based backpropagation combines:
- (i) Hard logic and fuzzy logic
- [✓ (ii) Fuzzy reasoning with neural learning]
- (iii) Symbolic reasoning with rules
- (iv) None of the above

(h) Crossover in GA is used to:
- (i) Randomize the population
- [✓ (ii) Exchange genetic material between parents]
- (iii) Delete unfit chromosomes
- (iv) Compute fitness

(i) Mutation in GA helps in:
- (i) Enhancing selection pressure
- [✓ (ii) Maintaining diversity in the population]
- (iii) Reducing fitness
- (iv) Stopping convergence

(j) The Fuzzy Logic Toolbox in MATLAB is used for:
- (i) Solving differential equations
- (ii) Designing fuzzy inference systems
- (iii) Generating random numbers
- (iv) Creating neural networks

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Q.2 (a) Explain the evolution of computing from conventional computing to soft computing [7]

(b) Illustrate the relationship between Artificial Intelligence, Machine Learning, and Soft Computing. [7]

Q.3 (a) Discuss the learning process in neural networks and the concept of weights and biases. [7]

(b) Discuss the difference between supervised and unsupervised learning in neural networks. [7]

Q.4 (a) Explain the concept of reinforcement learning in the context of neural networks. [7]

(b) Differentiate between supervised, unsupervised, and reinforcement learning approaches. [7]

Q.5 (a) Explain the concept of Fuzzy Sets and discuss how they differ from classical (crisp) sets. Illustrate your answer with suitable examples. [7]

(b) Describe the various operations on Fuzzy Sets such as union, intersection, complement, and Cartesian product. Provide appropriate examples. [7]

Q.6 (a) Describe the architecture and learning process of a Fuzzy-based Backpropagation Neural Network. How does it combine the advantages of fuzzy logic and neural networks? [7]

(b) Explain the basic concept and working principle of a Genetic Algorithm (GA) with a neat flowchart. [7]

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Q.7 (a) Discuss how Genetic Algorithms are used for optimization in Machine Learning models. Provide suitable examples. [7]

(b) Explain the concept of knowledge acquisition in Machine Learning and how Genetic Algorithms facilitate this process. [7]

Q.8 (a) Explain the architecture and working of a GA-based Backpropagation Neural Network (GA-BPNN). [7]

(b) Describe the working environment and structure of MATLAB and Python programming. [7]

Q.9 (a) Explain the use of NumPy arrays in Python and their advantages over traditional lists. [7]

(b) Explain the purpose and capabilities of the Neural Network Toolbox in MATLAB. [7]

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